

# Supplementary Material: muSync-GS

## Anonymous submission

Appendix A defines the evaluation protocols, Appendix B provides the complete vehicle-state update, Appendix C reports the CarSim and A2D2 evaluations, Appendix D presents additional video and visual results, and Appendix E reports the bounded surface-parameter sensitivity.

### A Evaluation Protocols and Reproducibility

#### A.1 Experiment map

Table 1 separates the 19-case calibration set from the frozen four-case speed check and 12-case OOD suite. Video and visual evaluations do not modify the vehicle model.

#### A.2 CarSim signal comparison

CarSim comparisons use original time and body pitch including the static offset; only muSync-GS is sampled at CarSim timestamps. We apply no time, path-position, or event alignment, pitch subtraction, residual correction, or case deletion. Slip is scored above 0.5 m/s, and wheel-level slip/load RMSEs are macro-averaged.

For flat-road braking, the event window begins at brake application and ends one second after the CarSim stop. For road-elevation-profile cases, it begins 0.5 m before front-axle feature entry and ends 0.5 m after rear-axle exit. The event metric is stopping-distance absolute error for flat-road cases and feature-center speed absolute error for road-profile cases.

#### A.3 Video-baseline conditioning

We compare Cosmos (NVIDIA 2025), LTX-Video (HaCohen et al. 2024), and VACE (Jiang et al. 2025). We match the semantic target condition, source clip, 81-frame output length, and seeds 2025–2027 while retaining each generator’s native conditioning interface. Table 2 lists the model versions, principal inputs, prompts, and frozen inference settings used for the submitted results.

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Prompt matching follows the same native-interface split. For W and O, all three generators receive the identical condition-level positive prompt. For G, Cosmos and LTX receive the identical prompt, while VACE uses a semantically matched mask-native prompt that requests the same feature class inside the “marked road region” without metric dimensions. No prompt supplies a CarSim trajectory, a metric height-versus-station trace, a response amplitude, or a target camera pose.

The shared positive-prompt structure is given below; quoted clauses are verbatim, with bracketed text replaced by the condition description. W uses: “A realistic forward-facing dashcam video of the same passenger vehicle braking on the same suburban residential street. Preserve the supplied road geometry, camera viewpoint, vehicle trajectory, braking motion, scene layout, houses, parked cars, and lane markings. Transform the global appearance to [weather description].” G first requests the same sunny street, viewpoint, forward trajectory, scene content, and temporal continuity. The Cosmos/LTX flat control states that the asphalt remains completely flat and forbids a hump, depression, pothole, trench, ramp, or other obstacle. Their feature clause requests either a 0.60 m-long, 0.10 m-high asphalt speed hump or a 2.00 m-long, 0.145 m-deep smooth road depression fixed at the supplied road location. O uses: “Faithfully preserve the supplied scene geometry, camera trajectory, braking and suspension motion,” followed by the matching weather and road description. VACE instead requests the corresponding hump or depression only inside its supplied mask.

The G protocol is a semantic geometry-edit diagnostic rather than an exact-profile accuracy comparison. The Cosmos/LTX prompts specify the dimensions above, while VACE receives only the semantic feature type and mask. The CarSim reference uses a  $0.075\text{ m} \times 3.7\text{ m}$  hump and a  $0.080\text{ m} \times 2.0\text{ m}$  depression with raised-cosine profiles. Generator inputs do not receive those height-versus-station traces. Consequently, the CarSim-referenced NRMSE tests whether an event-local pitch response appears with the expected sign and timing; it must not be read as an exact same-profile accuracy ranking. Cosmos uses a fixed negative prompt covering cartoon/CGI appearance, low quality, temporal flicker, warped roads, moving road obstacles, duplicate lane markings, camera cuts or new viewpoints, and text overlays. LTX uses a fixed negative prompt covering camera cuts,

| Component                        | Evaluation set                                                                                                                                                                                     | Primary comparison                                                                                                                                       | Parameter status              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Vehicle-model calibration        | 19 CarSim development cases: 9 flat braking, 5 speed humps, and 5 road depressions                                                                                                                 | Original-time speed, body pitch, slip, normal load, and event outcomes                                                                                   | Calibrated on development set |
| Initial-speed check              | 4 road-feature cases at unseen initial speeds                                                                                                                                                      | Same CarSim state and event metrics                                                                                                                      | Frozen                        |
| Main CarSim OOD                  | 12 cases: 5 weather-severity, 3 brake-command, and 4 road-elevation-profile holdouts                                                                                                               | CarSim state RMSE and stopping-distance or feature-center error                                                                                          | Frozen                        |
| External real-vehicle evaluation | 14 retained braking events from 3 A2D2 Audi Q7 drives                                                                                                                                              | Brake-pressure-driven pitch angle and pitch rate over the first 0.5 s                                                                                    | Leave one complete drive out  |
| Video-response protocols         | 8 non-sunny weather conditions, 2 geometric-road-hazard tasks, 2 deterministic geometry-only rendered CarSim state-reference clips, and 9 motion-preservation clips; generator results use 3 seeds | VGGT-recovered video response against CarSim, input-output preservation, and an empirical rendering/VGGT recovery reference, plus lateral path deviation | Frozen                        |
| Visual control                   | 9 weather conditions (5 displayed) and paired speed-hump/road-depression edits                                                                                                                     | Matched-frame appearance, quality/severity, localized edit, road-anchor, and temporal-mask metrics                                                       | Frozen                        |

Table 1: Map of the experiment sets. “Frozen” means that no calibration parameter is updated using the listed evaluation.

route, viewpoint, lane-layout or vehicle-position changes, duplicate vehicles, distorted roads, blur, and jitter. VACE uses the default negative prompt in its official inference script. Condition references are fixed across seeds within each task. Where a source-derived context frame is used, its index is 28; LTX O uses RGB only.

Cosmos and LTX use  $480 \times 320$  RGB streams; Cosmos additionally receives source depth. VACE’s 480p interface produces  $768 \times 512$  controls and outputs, which are canonicalized to H.264 at  $480 \times 320$  for evaluation. Every evaluated sequence contains 81 frames. W and O use 10 fps; the baseline geometry clips retain their common 16 fps protocol, while muSync-GS and the rendered CarSim state reference use 10 fps. Geometry scores are computed on the fixed 2.2–3.8 s clip-time interval with CarSim sampled on the corresponding fixed grid. No score uses content-dependent frame selection or a method-specific temporal shift.

#### A.4 muSync-GS rendering implementation

WeatherEdit (Qian et al. 2026) supplies the static atmospheric transform and Gaussian-particle renders. For every ego pose, paired renders with and without particles define the signed display-space residual

$$\Delta I_{\text{part}}(t) = I_{\text{with-part}}(t) - I_{\text{no-part}}(t), \quad (1)$$

which is added after pose synchronization. Particle count is a monotonic function of precipitation rate, while road accumulation is controlled separately by the surface state.

Rain uses semantic-road, RGB, depth, and normal buffers with reflection, refraction, and grazing-angle wet-road compositing following the organization of RainyGS (Dai et al. 2025). Exact visual anchors are the WeatherEdit wet-road baseline at 2 mm/h, wet sheen at 10 mm/h, and progressively stronger static thin-film composites at 25 and 50 mm/h. An intermediate rate linearly blends the adjacent pose-matched static anchors; its filtered particle residual

uses the analogous weight from the continuous target particle count. Dynamic shallow-water ripples, impacts, and splashes are disabled in the reported videos.

Snow adapts the normal-guided accumulation principle of Weather-Magician (Sang et al. 2025). A Gaussian is retained as a road anchor only if it is registered to the reconstructed semantic road, lies within the stored signed-height tolerance, and satisfies  $\mathbf{n}_i^\top \mathbf{u} \geq \cos 30^\circ$ . Normals use the shortest Gaussian axis when reliable and otherwise local PCA. A seed-conditioned deterministic coarse-to-fine field  $\xi_i \in [0, 1]$  activates anchor  $i$  when  $\xi_i \leq c_s$ , producing nested support as coverage increases. Local neighbor planes are footprint-filtered and offset by target snow depth plus micro-height variation. The same seed and scene assets reproduce the same coverage field.

#### A.5 Video-motion protocols and metrics

All weather methods use visual seeds 2025–2027. In muSync-GS, only particles and spatial snow realization vary; precipitation/depth controls, friction, telemetry, station, and camera poses remain bitwise frozen. Road geometry is deterministic (*det.*). Every clip uses the same camera-only VGGT procedure (Wang et al. 2025). Three protocols separate response generation from preservation. Because the W prompt explicitly preserves source trajectory and braking motion, W is a prompt-constrained appearance-editing diagnostic, not a fair ranking of missing physical-motion generation:

- **Weather-motion diagnostic (W):** a sunny braking input measures the residual difference from the condition-specific CarSim response while the generator is instructed to preserve source motion.
- **Geometry-response diagnostic (G):** every method receives the same sunny flat-road source and tests whether a speed-hump or road-depression edit creates a corresponding camera-pitch event; this is semantic rather than exact-profile accuracy.

- **Motion-preservation protocol:** an input with a condition-specific trajectory tests whether supplied event motion survives generation.

The O set contains  $\{\text{sunny, Rain-50, Snow-3}\} \times \{\text{braking, hump, depression}\}$ , for nine clips per seed. Only its six road-event clips enter the event-local pitch-correlation diagnostic; the three flat-road braking clips remain part of the preservation set but do not contain a localized road-pitch event.

Because monocular translation has unknown scale, forward progress is normalized independently for each sequence. Let  $\hat{p}_c(t)$  and  $\hat{p}_0(t)$  denote VGGT-recovered normalized progress for condition  $c$  and its sunny control, and let  $p_c^{\text{CS}}(t)$  and  $p_0^{\text{CS}}(t)$  denote the matched CarSim quantities. On the fixed  $T = 8$  s window, raw recovered forward coordinate is resampled to the common 81-frame grid and smoothed before terminal-direction sign correction and min-max normalization to  $[0, 1]$ . Normalized progress is smoothed once more, monotonized by a cumulative maximum, and clipped to  $[0, 1]$ . CarSim station is sampled at  $t = 0, 0.1, \dots, 8.0$  s, shifted to start at zero, and divided by its maximum over the same support. The same processing applies to the sunny control. The prompt-constrained W error is

$$E_{\text{weather}}(c) = \text{RMSE}[(\hat{p}_c - \hat{p}_0), (p_c^{\text{CS}} - p_0^{\text{CS}})]. \quad (2)$$

Condition errors are averaged within each seed before the reported mean and sample standard deviation (ddof = 1) are computed across seeds. Because conditioning requires motion preservation, this value measures distance from a counterfactual physical reference under that constraint; it is reported as a diagnostic and not interpreted as generator physics accuracy.

For muSync-GS, a replay check compares every seeded output with the reference physical rollout. The branch-wise audit has 30 runs: five rain-branch conditions (including sunny) and five snow-branch conditions (including sunny), each over three seeds. The visual-quality set contains 27 unique videos because the two branch-wise sunny controls are deduplicated. Across the 30 audit runs, the maximum absolute difference is zero for all 19 logged physical and pose-driving channels, including precipitation/depth controls, effective friction, speed, station, pitch, heave, slip, road excitation, brake pressure, and source-frame position. Thus, the reported variation isolates visual stochasticity rather than rerunning or perturbing the vehicle response.

The pose-matched particle residual is computed from paired renders decoded in 8-bit display-color space. It is added to the pose-corrected static render and clipped to the valid  $[0, 255]$  intensity range before encoding.

**Artifact and seed release.** Upon publication, we will release the implementation, preprocessing and evaluation scripts, frozen configurations, run manifests, and all random seeds under a license permitting free research use. Each stochastic result will be paired with its configuration and seed, including seeds 2025–2027 for visual generation and seed 20260725 for surface sensitivity; deterministic runs will be marked explicitly.

**Computing infrastructure.** The Linux-side reconstruction, Gaussian rendering, and video evaluation were run on a workstation with two NVIDIA RTX PRO 6000 Blackwell Max-Q Workstation Edition GPUs (96 GB each), an Intel Xeon w9-3575X CPU (44 physical cores and 88 threads), and 128 GB of system RAM under Ubuntu 24.04.2 LTS. The Gaussian stack used Python 3.10.19 and PyTorch 2.9.1 with CUDA 12.8; the VGGT and VACE environments used Python 3.10.20 and PyTorch 2.8.0 with CUDA 12.8. Reference vehicle-dynamics traces were generated with CarSim 2022.1. The planned artifact release will include environment files and manifests recording task-specific dependencies and model checkpoints.

**Rendered CarSim state reference.** For each geometry case, we render the same reconstructed scene and road asset with the original pose-neutral camera route, then apply the exact matched CarSim speed, station, body-pitch, and CG-height channel  $Z_{\text{cg}}$ . The reported pitch channel is total CarSim body pitch, while the pose update applies only  $\theta_{\text{CS}}(t) - \theta_0$  with  $\theta_0 = 0.154150^\circ$ ; the static orientation is therefore not applied twice. Dynamic heave is  $Z_{\text{cg}} - \text{median}(Z_{\text{cg}}[t < 0.8 \text{ s}])$ . The full exports contain 10,001 samples at 0.001 s intervals; before rendering, their speed, station, and pitch are required to match the frozen main-paper Table 3 CarSim CSVs within  $10^{-7}$ , and the observed maximum differences are zero for both cases. We render 81 frames at 10 fps and apply the same VGGT camera-only estimator, fixed global pitch sign, event window, and baseline flanks used for every geometry result, without temporal shift or sign/amplitude fitting. The rendered clips are then encoded with the same H.264,  $480 \times 320$ , 81-frame, 10-fps settings used by the main-paper Table 3 muSync-GS inputs. The resulting hump/depression NRMSEs are 0.354/0.406. They differ from muSync-GS by at most 0.010, providing an empirical rendering and monocular-recovery reference rather than a strict mathematical lower bound. This rendered-video reference is distinct from the CarSim state-space reference in the ablation, which bypasses rendering and VGGT.

For geometry response, the physical event window is fixed to 2.2–3.8 s. A local linear baseline is estimated from the fixed flanking intervals  $[1.70, 2.15]$  and  $[3.85, 4.30]$  s. Generator responses use paired feature-minus-flat pitch; muSync-GS and the rendered CarSim state reference use their respective single rendered road clips. Thus the column is an end-to-end semantic event-response diagnostic, not a common matched-profile estimator. All VGGT pitch curves use the same 11-frame, second-order Savitzky–Golay smoothing before the local baseline fit. This corresponds to 0.6875 s at 16 fps for generator G and 1.1 s at 10 fps for muSync-GS/state-reference clips; the unmatched duration is another reason not to interpret the result as exact-profile accuracy. The CarSim reference is sampled directly on the corresponding fixed 10- or 16-fps grid and is neither smoothed nor shifted. The VGGT camera-pitch sign is converted once to the CarSim vehicle-pitch convention. The normalized error is

$$E_{\text{geometry}} = \frac{\text{RMSE}(\hat{\theta}, \theta^{\text{CS}})}{\text{RMS}(\theta^{\text{CS}})}, \quad (3)$$

so a zero predicted event response gives  $E_{\text{geometry}} = 1$ . No method-specific temporal shift, sign fitting, or amplitude fitting is permitted. For each O weather, the matching braking pitch is subtracted from the hump/depression pitch separately for the input and output; each difference is linearly detrended using frames outside 16–42. Source-motion retention is the absolute input–output Pearson correlation on frames 16–42 (1.6–4.2 s at 10 fps), averaged over the six road-event clips within each seed and then across seeds. Higher correlation indicates stronger retention of motion already supplied in the O-protocol input; it is a preservation diagnostic rather than a counterfactual-generation score. It is not reported for muSync-GS because the method does not take the O-protocol video as an editor input, so no like-for-like input–output pair exists.

## A.6 Visual-control metrics

Weather appearance is compared at the same source scene, video time, and seed; it is not used as evidence of correct vehicle response. Road-geometry metrics use each geometry-response feature video and its paired flat control. Frames are first registered by background-only ECC affine motion, after which a paired residual mask is extracted inside a broad road region:

- **Target-localized edit rate** is the fraction of target-visible frames with a detectable paired change in the prescribed road region. It measures localization, not semantic realism.
- **Road-anchor error** is the distance between detected and prescribed mask centroids, normalized by image diagonal. Missing detections receive an error of one.
- **Temporal mask IoU (T-IoU)** warps the preceding edit mask to the next frame using optical flow. Missing masks count as zero.

The binary target mask uses grayscale threshold 127 and is active at  $\geq 80$  pixels. ECC affine registration excludes an 18-pixel target dilation and the lower 12% of the frame, with 70 iterations, tolerance  $10^{-5}$ , and Gaussian filter size 5. The paired residual is mean absolute Lab difference, thresholded at  $\max[0.025, \text{median}_{\text{bg}} + 3.5 \max(\text{MAD}_{\text{bg}}, 0.004)]$  inside the target-expanded lower-central road search region. A  $3 \times 3$  opening and  $7 \times 5$  closing are applied; a detection requires a connected component of at least 20 pixels overlapping the expanded target and at least 5% target coverage. T-IoU uses Farneback backward flow with pyramid scale 0.5, four levels, window 21, four iterations, `poly_n` = 7, and `poly_sigma` = 1.5. Metrics are averaged over active frames within a clip, then over the two road tasks within each seed; reported generator uncertainty is the three-seed sample SD (ddof = 1).

Background LPIPS is excluded from the primary road table because global weather transfer and the intended physics-aware viewpoint response can legitimately change pixels outside a local road mask. For camera-path preservation, each VGGT trajectory is independently normalized by its forward extent and interpolated with its sunny counterpart on a common normalized-progress grid  $u \in [0, 1]$ . If  $\tilde{y}_c(u)$  and  $\tilde{y}_0(u)$

are their normalized lateral coordinates, the clip error is

$$E_{\text{path}}(c) = \text{RMS}_{u \in [0,1]} [\tilde{y}_c(u) - \tilde{y}_0(u)]. \quad (4)$$

This prevents different braking durations from being counted as lateral drift. For every method, the eight non-sunny conditions are macro-averaged within each seed, followed by the mean and sample SD across seeds.

## B Complete Vehicle-State Update

This section gives the update rules omitted from the compact main-paper description. The reported implementation uses axle-level wheel states for straight-line motion; the corresponding left/right outputs are identical and each reported per-wheel load is one half of its axle load. Its continuous state is

$$\mathbf{x} = \left( v, s, \omega_f, \omega_r, z_s, \dot{z}_s, \theta, \dot{\theta}, z_{u,f}, \dot{z}_{u,f}, z_{u,r}, \dot{z}_{u,r} \right). \quad (5)$$

The initial conditions are  $v(0) = v_0$ ,  $s(0) = 0$ ,  $\omega_f(0) = \omega_r(0) = v_0/R$ , and zero half-car perturbation states and velocities. The ABS torque scales are initialized to one. The half-car coordinate  $\theta(t)$  is therefore a dynamic pitch perturbation. The CarSim state comparison reports  $\theta_{\text{body}}(t) = \theta_0 + \theta(t)$  with the frozen static offset  $\theta_0 = 0.154150^\circ$ , whereas the camera-pose composition applies only  $\theta(t)$  to the already reconstructed reference pose. Thus, the static orientation is not applied twice. The scenario supplies the precipitation-conditioned surface coefficient and brake-pressure command  $p_b(t)$ . Here  $s(t)$  denotes the front-axle route coordinate, with  $\dot{s} = v$ ,  $z_{r,f} = H_g(s)$ , and  $z_{r,r} = H_g(s - L_w)$ . The total modeled mass is  $m = m_s + m_{u,f} + m_{u,r}$ .

### B.1 Reference surface coefficient

The CarSim comparisons use the precipitation-derived coefficient  $\mu_c$  defined in the main paper. Flat-road weather and brake cases start at 60 km/h, road-profile cases at 40 km/h, and the separate road-profile speed check at 35/45 km/h. The frozen  $\eta_{\text{tire}}$  and  $\beta_\kappa$  values are selected from  $\mu_c$  and held fixed through each stop.

### B.2 Wheel, brake, and ABS update

For axle  $a \in \{f, r\}$ , the kinematic braking slip and horizontally scaled tire-curve coordinate are

$$\kappa_a = \text{clip} \left( \frac{v - R\omega_a}{\max(v, v_\epsilon)}, 0, 1 \right), \quad \tilde{\kappa}_a = \beta_{\kappa,w} \kappa_a, \quad (6)$$

where  $v_\epsilon = 0.5 \text{ m/s}$  is the slip denominator floor and  $\beta_{\kappa,w}$  is interpolated within the weather family. The longitudinal force is

$$F_{x,a} = \mu_B(\tilde{\kappa}_a; \mu_{\text{eff}}) F_{z,a}. \quad (7)$$

Let

$$b(\kappa) = c_1(1 - e^{-c_2\kappa}) - c_3\kappa, \quad (8)$$

$$\mu_B(\kappa; \mu_{\text{eff}}) = \frac{\mu_{\text{eff}}}{\max_{\lambda \in [0,1]} b(\lambda)} b(\kappa).$$

This is the peak-normalized Burckhardt curve (Burckhardt 1993). The  $(c_1, c_2, c_3)$  tuples are (1.2801, 23.99, 0.52) for

| Method                | W input                                                  | G input                                                                          | O input                                        | Frozen inference settings                                                               |
|-----------------------|----------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------|
| Cosmos-Transfer2.5-2B | Sunny RGB, source depth, and a condition reference image | Flat RGB, source depth, local depth-application mask, and flat/feature reference | Condition-specific and depth; context frame 28 | Official depth checkpoint; 35 steps, guidance 3; depth weights 0.75/0.70/0.90 for W/G/O |
| LTX-Video 13B 0.9.8   | Sunny RGB and condition reference; no depth or mask      | Flat RGB and flat/feature reference at frame 28; no depth or mask                | Condition-specific only                        | Fixed prompt without enhancement; 30 steps; initial-step skips 17 for W/G and 24 for O  |
| Wan2.1-VACE-14B       | Depth-conditioned sunny video and condition reference    | Flat RGB, local edit mask, and flat/feature reference                            | Condition-specific video and reference         | 50 UniPC steps, guidance 5, shift 16; plain prompt mode                                 |

Table 2: Native conditioning and frozen generation settings for the video baselines. W, G, and O denote the prompt-constrained weather-motion diagnostic, semantic geometry-response diagnostic, and motion preservation, respectively.

dry asphalt, (0.857, 33.822, 0.347) for wet asphalt, and (0.1946, 94.129, 0.0646) for snow. Scaling changes the peak to  $\mu_{\text{eff}}$  without changing the base peak-slip location. Its target slip is  $\kappa_a^* = \arg \max_{\kappa} \mu_B(\beta_{\kappa, w} \kappa; \mu_{\text{eff}})$ . The pressure command is a scenario-defined ramp/hold/release profile. It is mapped to axle torque by

$$T_b^{\text{cmd}} = K_p p_b(t), \quad T_{b,a} = b_a q_a T_b^{\text{cmd}}, \quad (9)$$

with  $K_p = 1000 \text{ N m/MPa}$ ,  $b_f = 0.65$ , and  $b_r = 0.35$ . The formal controller uses the proportional ABS command

$$q_a^{\text{cmd}} = \text{clip}\left(\frac{\kappa_a^*(1 + \delta_a) - \kappa_a}{\delta_a \kappa_a^*}, 0, 1\right), \quad (10)$$

where  $\kappa_a^*$  is the frozen target slip and  $\delta_a = 0.2$ . The reported configuration uses algebraic pressure response,  $q_a = q_a^{\text{cmd}}$ ; the implementation also exposes the more general rate-limited update

$$\begin{aligned} \Delta q_a^n &= \text{clip}(q_a^{\text{cmd}} - q_a^n, -r_{\text{rel}} \Delta t, r_{\text{build}} \Delta t), \\ q_a^{n+1} &= \text{clip}(q_a^n + \Delta q_a^n, 0, 1), \end{aligned} \quad (11)$$

whose build and release rates are set sufficiently large to recover the algebraic rule in all reported runs. Wheel rotation, longitudinal speed, and path position obey

$$\begin{aligned} I_{w,a} \dot{\omega}_a &= R F_{x,a} - T_{b,a}, \\ m \dot{v} &= -(F_{x,f} + F_{x,r}) + F_{x,\text{road}} - m a_{\text{coast}}, \\ F_{x,\text{road}} &= -F_{z,f} H'_g(s) - F_{z,r} H'_g(s - L_w), \\ \dot{s} &= v. \end{aligned} \quad (12)$$

At the stopping threshold  $v \leq 0.05 \text{ m/s}$ , the rollout sets  $v = \omega_f = \omega_r = \kappa_f = \kappa_r = 0$  and resets the ABS torque scales.

### B.3 Four-DOF half-car and contact forces

The half-car coordinates are  $\mathbf{x}_h = (z_s, \theta, z_{u,f}, z_{u,r})$ . Define

$$\begin{aligned} d_f &= z_s - z_{u,f} - l_f \theta, & F_{s,f} &= k_{s,f} d_f + c_{s,f} \dot{d}_f, \\ d_r &= z_s - z_{u,r} + l_r \theta, & F_{s,r} &= k_{s,r} d_r + c_{s,r} \dot{d}_r. \end{aligned} \quad (13)$$

Because the coordinates are perturbations about static equilibrium, the static axle loads are

$$F_{z,f}^0 = m_s g \frac{l_r}{L_w} + m_{u,f} g, \quad F_{z,r}^0 = m_s g \frac{l_f}{L_w} + m_{u,r} g. \quad (14)$$

The tire-force increments and actual unilateral contact loads are

$$\begin{aligned} F_{t,a} &= \max(k_{t,a}(z_{r,a} - z_{u,a}), -F_{z,a}^0), \\ F_{z,a} &= F_{z,a}^0 + F_{t,a}. \end{aligned} \quad (15)$$

The complete four-equation model is

$$\begin{aligned} m_s \ddot{z}_s &= -F_{s,f} - F_{s,r}, \\ I_y \ddot{\theta} &= l_f F_{s,f} - l_r F_{s,r} + M_b, \\ m_{u,f} \ddot{z}_{u,f} &= F_{s,f} + F_{t,f}, \\ m_{u,r} \ddot{z}_{u,r} &= F_{s,r} + F_{t,r}, \end{aligned} \quad (16)$$

with braking moment

$$M_b = \gamma_{\theta, w} m_s h_{\text{cg}} d_{\theta}(t). \quad (17)$$

Away from the stopping tail,  $d_{\theta}(t) = \max(-\dot{v}, 0)$ . The half-car is integrated by RK4 using linearly interpolated front and rear road inputs.

### B.4 Low-speed pitch rule and coupled update order

The low-speed rule modifies only the brake-pitch input in Eq. 17; it does not modify simulated speed or stopping distance. Let  $t_s$  be the first sample with  $v \leq 0.05 \text{ m/s}$  and  $t_h = \max(t_b, t_s - \tau_h)$ , where  $t_b$  is brake onset and  $\tau_h$  is the frozen family-dependent hold duration. We compute a reference deceleration  $\bar{d}_h$  as the median positive deceleration over the preceding 0.6 s. If the median over the final 0.1 s before  $t_s$  is below  $\rho_h \bar{d}_h$ , then  $d_{\theta}(t) = \bar{d}_h$  on  $[t_h, t_s]$ ; otherwise the final 0.2 s is capped at  $1.05 \bar{d}_h$ . After stopping, the held value decays linearly to zero over the frozen release duration  $\tau_r$ . The reported configuration uses  $\rho_h = 0.8$ ; all  $\tau_h$  and  $\tau_r$  values are listed in Table 5.

Road-feature cases use the same half-car equations and parameter set for road and braking inputs. The implementation evaluates

$$\mathbf{x}_h = \mathbf{x}_{h,\text{road}} + \mathbf{x}_{h,\text{brake}}. \quad (18)$$

All evaluated cases remain in the active-contact regime, so the unilateral clamp in Eq. 15 is inactive and modal superposition is equivalent to applying both inputs to the common linear system. The combined state supplies both axle loads and rendered camera heave/pitch. For road-feature cases, the

solver performs five global trajectory-level fixed-point passes over the full internal time grid, not five local iterations per output sample. Each pass:

1. evaluate  $z_{r,f/r}$  from the current path position;
2. integrate the common half-car and compute  $F_{z,f/r}$ ;
3. update tire force, wheel rotation, speed, and path position using Eqs. 10–12.

Flat-road braking uses acceleration-dependent longitudinal load transfer in the wheel solver,

$$\begin{aligned} F_{z,f}^{\text{long}} &= \max\left(m \frac{gl_r - \dot{v}h_{cg}}{L_w}, 0\right), \\ F_{z,r}^{\text{long}} &= \max\left(m \frac{gl_f + \dot{v}h_{cg}}{L_w}, 0\right). \end{aligned} \quad (19)$$

The common half-car separately provides the transient normal-load, heave, and pitch signals used for evaluation and rendering. Thus dynamic half-car loads feed the tire solver in road-feature cases, whereas flat-road longitudinal tire forces use the analytic load-transfer equation above. All mechanical values, calibrated scalars, and numerical step sizes are given in Tables 4 and 5.

## C CarSim Development and Frozen-Parameter OOD Results

### C.1 Calibration procedure and freezing

We distinguish specified mechanical values from calibration parameters. Mass, axle geometry, suspension coefficients, wheel properties, and the common effective tire stiffness are taken from the matched CarSim vehicle setup or its effective reduced-order representation (Table 4). The continuous calibration comprises 40 stored scalar entries, counted in Table 3. The eight  $\mu_c$  knot locations are fixed by the development surface conditions and are not counted as calibrated variables. Discrete structural choices—the common half-car/contact model, local contact-slope road force, and zero road-type-specific energy correction—were selected by development-stage structural ablations and then frozen; they are reported separately and excluded from the scalar count.

Calibration proceeded iteratively by response group using the original-time diagnostic vector

$$\mathbf{e}_{\text{cal}} = (\text{RMSE}_v, \text{RMSE}_\theta, \text{RMSE}_\kappa, \text{RMSE}_{F_z}, |\Delta d_{\text{stop}}|, |\Delta v_{\text{center}}|), \quad (20)$$

as the common evaluation record. Each parameter group was adjusted against its directly affected response as listed in Table 3. Normal-load RMSE was monitored throughout as a cross-check. The three structural switches were inspected in development-stage structural comparisons before the empirical entries were fixed; this section reports the selected structure rather than an additional quantitative ablation result. Slip stiffness was adjusted using per-wheel slip RMSE; longitudinal efficiency used flat-road speed and stopping-distance errors; and brake-pitch gains and low-speed hold/release parameters used body-pitch and stop-tail responses. The shared coupled pitch scale was subsequently checked on the road-feature cases. After each accepted update, all 19 development

| Calibration group                          |             | Scalars   | Response used during staged selection                                     |
|--------------------------------------------|-------------|-----------|---------------------------------------------------------------------------|
| Tire-force efficiency $\eta_{\text{tire}}$ | effi-       | 8         | Original-time speed and stopping-distance errors in flat braking          |
| Slip-stiffness $\beta_\kappa$              | scale       | 8         | Per-wheel slip-ratio RMSE                                                 |
| Brake-pitch $\gamma_\theta$                | scale       | 8         | Body-pitch RMSE during braking                                            |
| Low-speed hold/release and hold ratio      |             | 4 + 8 + 1 | Stop-tail and release portion of the body-pitch response                  |
| Coupled pitch scale                        | brake-      | 1         | Body-pitch RMSE in the ten road-feature development cases                 |
| Static offset                              | total-pitch | 1         | Initial total-pitch level read from the matched CarSim development traces |
| Coast-drag acceleration $a_{\text{coast}}$ | acceler-    | 1         | Shared development-frozen coasting response                               |
| Total                                      |             | 40        | All entries frozen before held-out evaluation                             |

Table 3: Accounting of the 40 scalar calibration entries. The third column records the diagnostic response used during staged selection.

cases were rerun and the speed, slip, pitch, normal-load, and event-level diagnostics were examined jointly. The final entries and their associated diagnostic responses are reported in Tables 3 and 5.

The 19 development cases comprise nine flat-road braking cases at 60 km/h—dry, rain 2/10/25/50 mm/h, and snow 0.5/1/2/3 mm/h—plus ten road-feature cases at 40 km/h. The latter apply five surface conditions (dry, rain 10, rain 50, snow 1, and snow 3) to each of a 75 mm  $\times$  3.7 m speed hump and an 80 mm  $\times$  2.0 m road depression. All use the specified 2.3 MPa brake command and 0.5 s pressure ramp for their respective protocol.

The four initial-speed cases in Table 6 are held out from calibration. The later 12-case OOD suite uses the same frozen configuration, with the four speed holdouts excluded from parameter selection. No learned or additive time-series residual correction is applied in the reported development or holdout results.

### C.2 OOD case manifest

The 12-case suite uses the calibration configuration listed above and changes three control families while retaining the reference vehicle and tire.

### C.3 Per-case OOD agreement

The road-geometry split gives the strongest state transfer. The largest errors are short weather- and brake-induced pitch transients: Rain 5, Snow 0.75, peak pressure 1.6 MPa, and ramp 0.2 s reach 0.0740–0.1021° pitch RMSE. These cases remain in every aggregate; their smaller speed and event errors indicate that the main discrepancy is transient brake

| Component              | Parameter                                                 | Value                                                     | Role/provenance                                                                                                         |
|------------------------|-----------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Mass and geometry      | $m_s$ ; $m_{u,f}, m_{u,r}$ ; $l_f, l_r$ ; CG height       | 1200 kg; 54/54 kg; 1.2/1.5 m; 0.55 m                      | Matched CarSim vehicle; total modeled mass is 1308 kg and wheelbase is 2.7 m.                                           |
| Sprung-body rotation   | $I_y$                                                     | 1800 kg m <sup>2</sup>                                    | Shared pitch inertia for braking and road excitation.                                                                   |
| Suspension             | $k_{s,f}, k_{s,r}$ ; $c_{s,f}, c_{s,r}$                   | 27,000/27,000 N/m; 3920/3920 N s/m                        | CarSim per-side values of 13.5 N/mm and 1960 N s/m combined per axle.                                                   |
| Tire contact           | $k_{t,f}, k_{t,r}$ ; contact constraint                   | 193,268.620855 N/m; unilateral                            | Shared effective axle tire stiffness; negative tire loads are disallowed.                                               |
| Wheel                  | $R$ ; rotational inertia                                  | 0.32 m; 2.4 kg m <sup>2</sup> per axle                    | Effective rolling and rotational parameters.                                                                            |
| Brake/ABS              | Pressure–torque gain; front bias; release width           | 1000 N m/MPa total; 0.65; 0.20                            | The common development command is 2.3 MPa with a 0.5 s ramp; OOD commands are listed separately.                        |
| Road coupling          | Longitudinal model; $a_{\text{coast}}$ ; road-energy gain | Local normal-load–slope force; 0.004 m/s <sup>2</sup> ; 0 | Local front/rear contact slopes; the shared coast term is counted in Table 3; no road-type-specific energy subtraction. |
| Integration/evaluation | Internal/output step; slip speed floor; duration          | 0.0005/0.001 s; 0.5 m/s; 10 s                             | Fixed across matched runs.                                                                                              |

Table 4: Fixed mechanical, controller, solver, and shared road-coupling settings in the final frozen configuration. Per-axle suspension values combine the two sides of the matched CarSim vehicle setup.

| Family | $\mu_c$  | $\eta_{\text{tire}}$ | $\beta_\kappa$ | $\gamma_\theta$ | $\tau_h$ (s) | $\tau_r$ (s) |
|--------|----------|----------------------|----------------|-----------------|--------------|--------------|
| Dry    | 0.820000 | 1.0100               | 1.65           | 1.104778        | 0.20         | 0.08         |
| Rain   | 0.347126 | 0.9278               | 1.00           | 1.133213        | 0.20         | 0.06         |
|        | 0.369279 | 0.9320               | 1.00           | 1.129315        | 0.20         | 0.06         |
|        | 0.434150 | 0.9427               | 1.00           | 1.123202        | 0.20         | 0.06         |
|        | 0.595585 | 1.0000               | 1.63           | 1.105687        | 0.20         | 0.08         |
| Snow   | 0.240000 | 0.9588               | 1.00           | 1.142519        | 0.20         | 0.06         |
|        | 0.420000 | 0.9932               | 1.00           | 1.111734        | 0.20         | 0.04         |
|        | 0.510000 | 0.9835               | 1.00           | 1.107972        | 0.05         | 0.06         |

Additional frozen scalars are static total pitch 0.154150°, low-speed hold ratio 0.8, and coupled brake-pitch scale 0.98. These terms reproduce the reported vehicle-response configuration. The displayed dry/rain  $\tau_h = 0.20$  s values share one stored scalar; together with three snow-specific values, the hold-time group therefore contains four rather than eight independent entries.

Table 5: Frozen calibration parameters. Within each weather family, the arrays are listed one row per ascending  $\mu_c$  knot and are interpolated piecewise linearly; values outside the knot range use the nearest endpoint.  $\eta_{\text{tire}}$  scales the surface coefficient,  $\beta_\kappa$  scales slip stiffness,  $\gamma_\theta$  scales brake-pitch moment, and  $\tau_h/\tau_r$  are low-speed hold/release durations. The knot locations are fixed and are not part of the 40-parameter count.

dive or release rather than accumulated stopping behavior. Complete trajectories following the evaluation rules in Appendix A.2 are shown in Figures 1– 3.

#### C.4 External A2D2 real-vehicle brake-dive evaluation

The A2D2 vehicle bus provides synchronized brake pressure, vehicle speed, pitch angle, pitch rate, and acceleration (Geyer et al. 2020). Pressure episodes are defined by samples at or above 1.2 bar; consecutive active samples separated by at most 0.08 s are merged. Of 140 detected episodes in three Audi Q7 drives, 138 support the full  $[-0.8, 1.0]$  s analysis window and 14 satisfy all criteria listed below. The three drives contribute 8, 2, and 4 retained events, with event-start speeds from 13.1 to 36.4 km/h. Pitch is expressed as nose-down change relative to a linear pre-event baseline, and the evaluation window covers the first 0.5 s after brake-pressure

onset.

#### A2D2 event-selection rules (138 complete-window candidates)

| Criterion                          | Threshold                   | Fails |
|------------------------------------|-----------------------------|-------|
| Active-pressure duration           | $\geq 0.5$ s                | 45    |
| Start speed                        | $\geq 12$ km/h              | 47    |
| Pressure increase                  | $\geq 3.0$ bar              | 44    |
| Deceleration increase within 0.5 s | $\geq 0.4$ m/s <sup>2</sup> | 41    |
| Absolute steering p95              | $\leq 10^\circ$             | 71    |
| Absolute yaw-rate p95              | $\leq 1.5^\circ/\text{s}$   | 48    |
| Absolute pre-event pitch slope     | $\leq 0.25^\circ/\text{s}$  | 64    |
| Pre-event pitch SD                 | $\leq 0.07^\circ$           | 11    |
| Onset pitch residual               | $\leq 0.12^\circ$           | 45    |
| Abs. vertical-acceleration p95     | $\leq 0.8$ m/s <sup>2</sup> | 8     |
| Absolute roll-rate p95             | $\leq 1.5^\circ/\text{s}$   | 40    |
| Pre-event pressure p95             | $\leq 0.9$ bar              | 51    |

| Control                        | Development values                 | Held-out values                            | Relationship                                                 |
|--------------------------------|------------------------------------|--------------------------------------------|--------------------------------------------------------------|
| Weather severity               | Rain 2/10/25/50;<br>0.5/1/2/3 mm/h | snow Rain 5/15/35; snow 0.75/1.5 mm/h      | Unseen within-family interpolation                           |
| Peak brake pressure            | 2.3 MPa                            | 1.6 MPa                                    | Lower off-grid command                                       |
| Pressure ramp                  | 0.5 s                              | 0.2 s                                      | Faster off-grid command                                      |
| Brake onset                    | Time trigger at 1.0 s              | Route-position trigger at 12 m             | Unseen trigger type                                          |
| Speed hump                     | 75 mm $\times$ 3.7 m               | 50 mm $\times$ 3.7 m; 75 mm $\times$ 2.5 m | Off-grid height or length                                    |
| Road depression                | 80 mm $\times$ 2.0 m               | 50 mm $\times$ 2.0 m; 80 mm $\times$ 3.0 m | Off-grid depth or length                                     |
| Initial speed (separate check) | 40 km/h for road cases             | 35/45 km/h                                 | Local bidirectional off-grid check; not in the 12-case suite |

Original development and initial-speed-check agreement (mean event-window RMSE).

| Set                  | Pitch ( $^{\circ}$ ) | Speed (m/s) | Slip    | Per-wheel $F_z$ (N) |
|----------------------|----------------------|-------------|---------|---------------------|
| Flat braking (9)     | 0.0475               | 0.0288      | 0.00958 | 16.77               |
| Speed hump (5)       | 0.0354               | 0.0171      | 0.00150 | 25.16               |
| Road depression (5)  | 0.0335               | 0.0185      | 0.00127 | 41.65               |
| All development (19) | 0.0406               | 0.0230      | 0.00527 | 25.53               |
| Speed holdout (4)    | 0.0263               | 0.0113      | 0.00011 | 22.35               |

For the nine development braking cases, braking-distance, braking-time, and slip-distance MAEs are 0.107 m, 0.0065 s, and 0.0535 m, respectively. The four initial-speed cases use the same frozen calibration and are not included in the 12-case suite.

Table 6: Development–evaluation control relationship and original development agreement. “Interpolation” is used only when held-out values lie between observed development levels within the same weather family. A singleton development setting cannot define an interpolation range, so the brake, geometry, and initial-speed changes are reported as off-grid tests.

| Held-out factor        | Cases                                                                                                          | Change relative to development protocol                                                                                                                             |
|------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Weather severity       | Rain 5/15/35 mm/h; snow 0.75/1.5 mm/h                                                                          | New within-family levels. The frozen mapping gives $\mu = 0.503/0.401/0.356$ for rain and $0.465/0.330$ for snow; the same value is passed to CarSim and muSync-GS. |
| Brake command          | Peak pressure 1.6 MPa; ramp 0.2 s; brake onset at 12 m along the route                                         | Changes magnitude, application rate, or start trigger while retaining the same surface, vehicle, tire, and calibration parameters.                                  |
| Road-elevation profile | Humps 50 mm $\times$ 3.7 m and 75 mm $\times$ 2.5 m; depressions 50 mm $\times$ 2.0 m and 80 mm $\times$ 3.0 m | Changes feature height or depth, length, and slope through the same signed raised-cosine interface.                                                                 |

Table 7: 12-case OOD design. All calibration parameters remain fixed during evaluation. Snow rates are water equivalent with the one-hour accumulation setting.

Failure counts overlap; p95 denotes the 95th percentile over  $[-0.2, 0.6]$  s around pressure onset.

Brake onset is the first sample with pressure at or above 1.2 bar. A common  $[-0.6, -0.1]$  s pre-event window supplies the median pressure and longitudinal acceleration base-lines and the affine pitch baseline. Pressure increase is the maximum nonnegative baseline-subtracted pressure over  $[0, 0.5]$  s; deceleration increase is the maximum negative longitudinal-acceleration change over the same window. All 12 selection criteria are applied conjunctively; no model prediction or pitch-agreement score enters event selection.

We use complete-drive leave-one-drive-out evaluation. For each fold, a shared set of effective vehicle parameters and the pressure gain are estimated using only the other two drives. Variants with fitted damping additionally estimate an effective damping ratio; the half-car, pressure-to-slip, tire, and controller equations are unchanged. Held-out inference receives the measured brake-pressure trace and event-start

speed only. Neither measured pitch nor  $\omega_y$  is supplied at test time. The main paper reports equal-drive macro statistics, while Figure 4 shows fold-level responses for every held-out drive and the angle-only calibration ablation.

The main-paper visualization is the drive-macro response over all retained held-out events, rather than a post-hoc selected event. All aggregate metrics and the fold-level pitch and pitch-rate summaries use the same 14 events and are reported in physical units.

Here, *calibration signals* are the measured signals from the two training drives used to estimate the fold-specific parameters; they are not provided as inputs for the held-out drive. “Pitch only” uses pitch angle as the calibration signal, whereas “Pitch + rate” jointly uses pitch angle and pitch rate. The damping column indicates whether the effective damping ratio is fixed at  $\zeta = 0.7$  or estimated from the training drives.

The pressure–torque gain is fit over 0.05–0.5 s by



| Split        | Case                           | Speed (m/s) | Pitch ( $^{\circ}$ ) | Slip     | Per-wheel<br>$F_z$ (N) | Event error  |
|--------------|--------------------------------|-------------|----------------------|----------|------------------------|--------------|
| Weather      | Rain 5                         | 0.09862     | 0.10208              | 0.036701 | 31.53                  | 0.28361 m    |
| Weather      | Rain 15                        | 0.02910     | 0.05128              | 0.019291 | 15.26                  | 0.14348 m    |
| Weather      | Rain 35                        | 0.02420     | 0.04858              | 0.018267 | 14.35                  | 0.11429 m    |
| Weather      | Snow 0.75                      | 0.05342     | 0.08417              | 0.013246 | 37.00                  | 0.14826 m    |
| Weather      | Snow 1.5                       | 0.01649     | 0.04342              | 0.016212 | 13.42                  | 0.01591 m    |
| Brake        | Peak pressure 1.6 MPa          | 0.01629     | 0.07403              | 0.005623 | 26.72                  | 0.02099 m    |
| Brake        | Ramp 0.2 s                     | 0.02406     | 0.09470              | 0.005945 | 36.04                  | 0.02028 m    |
| Brake        | Onset at 12 m along route      | 0.02372     | 0.05307              | 0.005553 | 21.99                  | 0.01353 m    |
| Road profile | Hump $50 \times 3700$ mm       | 0.00979     | 0.03031              | 0.000040 | 22.10                  | 0.03694 km/h |
| Road profile | Hump $75 \times 2500$ mm       | 0.01111     | 0.04135              | 0.000106 | 37.68                  | 0.04028 km/h |
| Road profile | Depression $50 \times 2000$ mm | 0.01068     | 0.03143              | 0.000062 | 28.71                  | 0.03878 km/h |
| Road profile | Depression $80 \times 3000$ mm | 0.01042     | 0.05318              | 0.000071 | 34.57                  | 0.04058 km/h |

Table 8: Per-case frozen-parameter OOD agreement. State columns are RMSE. The last column is stopping-distance absolute error for weather and brake cases and feature-center speed absolute error for road-geometry cases.

bounded robust least squares with soft- $L_1$  loss and bounds  $[10, 500]$  N m/bar. The pitch adapter is fit over 0–0.5 s with the same loss, natural-frequency bounds  $[0.5, 4.0]$  Hz, static-gain bounds  $[0.01, 1.2]$  degrees per  $\text{m/s}^2$ , and fixed damping ratio 0.70.

The main paper reports that joint pitch-angle-and-rate calibration attains drive-macro pitch RMSE  $0.062^{\circ}$  and correlation 0.901, with pitch-rate RMSE  $0.381^{\circ}/\text{s}$  and correlation 0.560. Per-drive agreement is not uniform: one fold underpredicts the late pitch amplitude, another overpredicts it, and pitch-rate timing remains less stable than pitch angle. These results support the direction and early evolution of real brake dive after vehicle-specific recalibration; they do not constitute zero-shot vehicle transfer, physical identification of the Audi Q7 suspension, or full-maneuver validation.

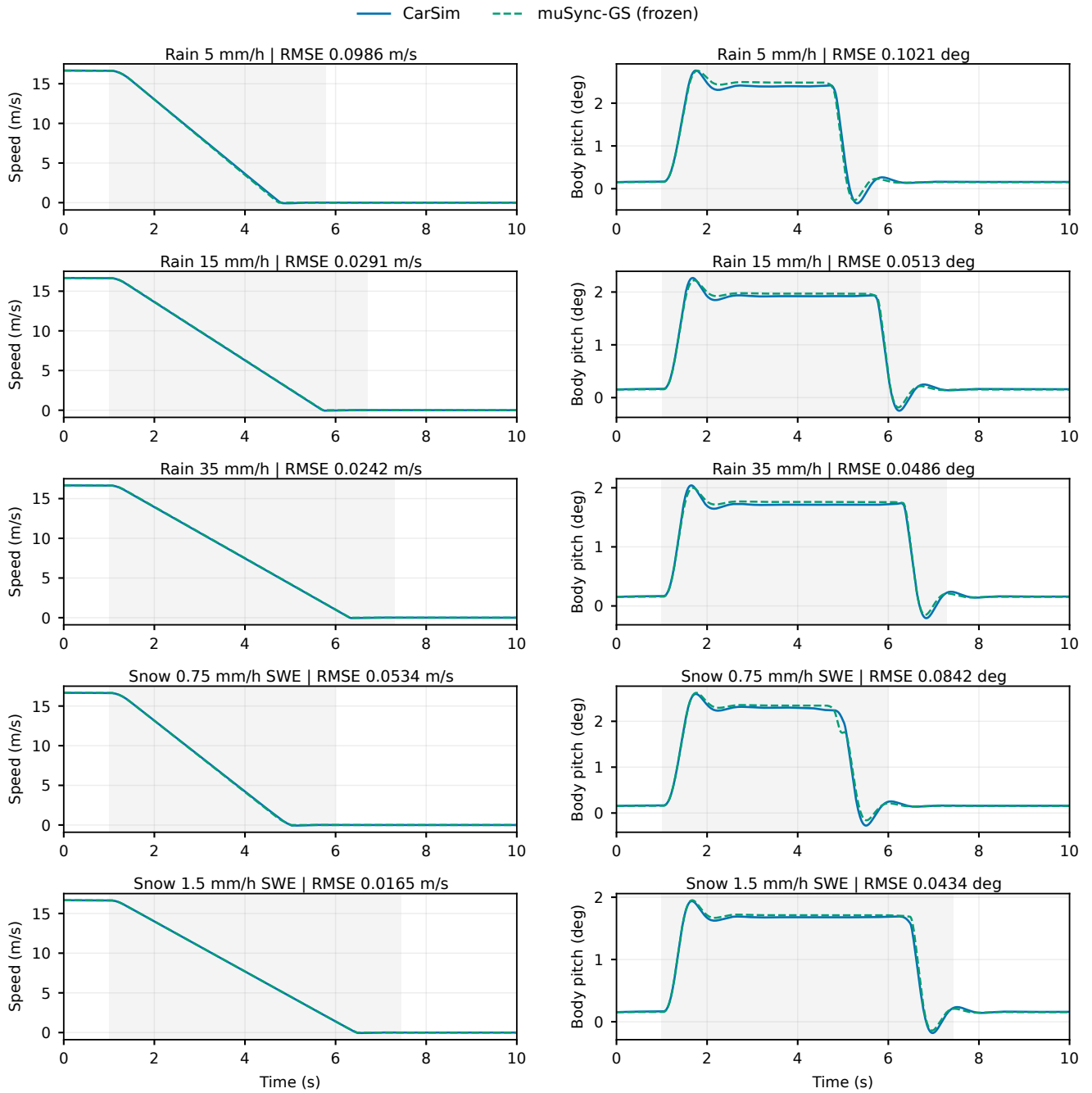


Figure 1: All five frozen weather-severity holdouts. Gray shading denotes the event window. Curves use original time and body pitch, including the static offset, without alignment or calibration updates.

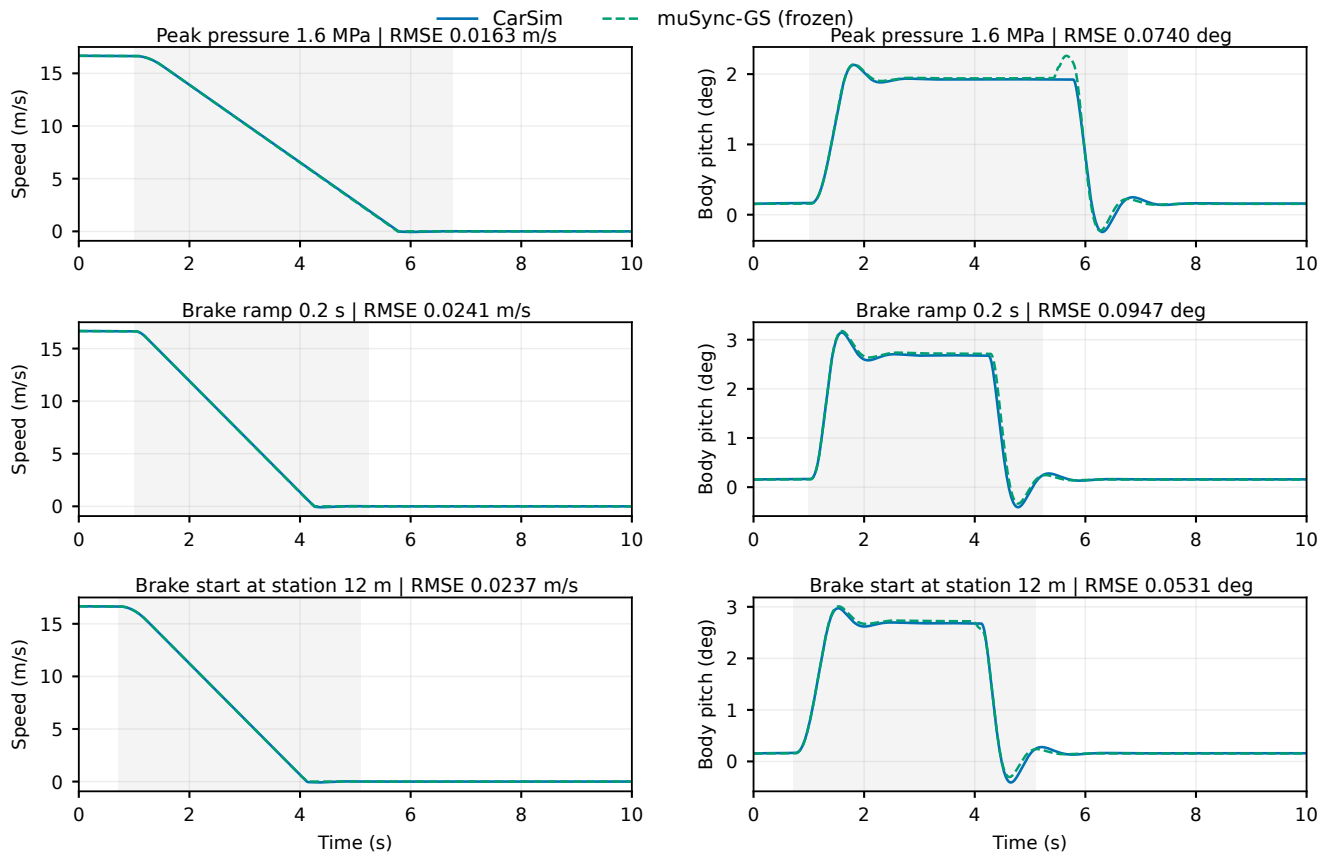


Figure 2: All three frozen brake-input holdouts. The 0.2 s ramp has the largest pitch transient error in this split, while stopping-distance error remains below 0.021 m in every case.

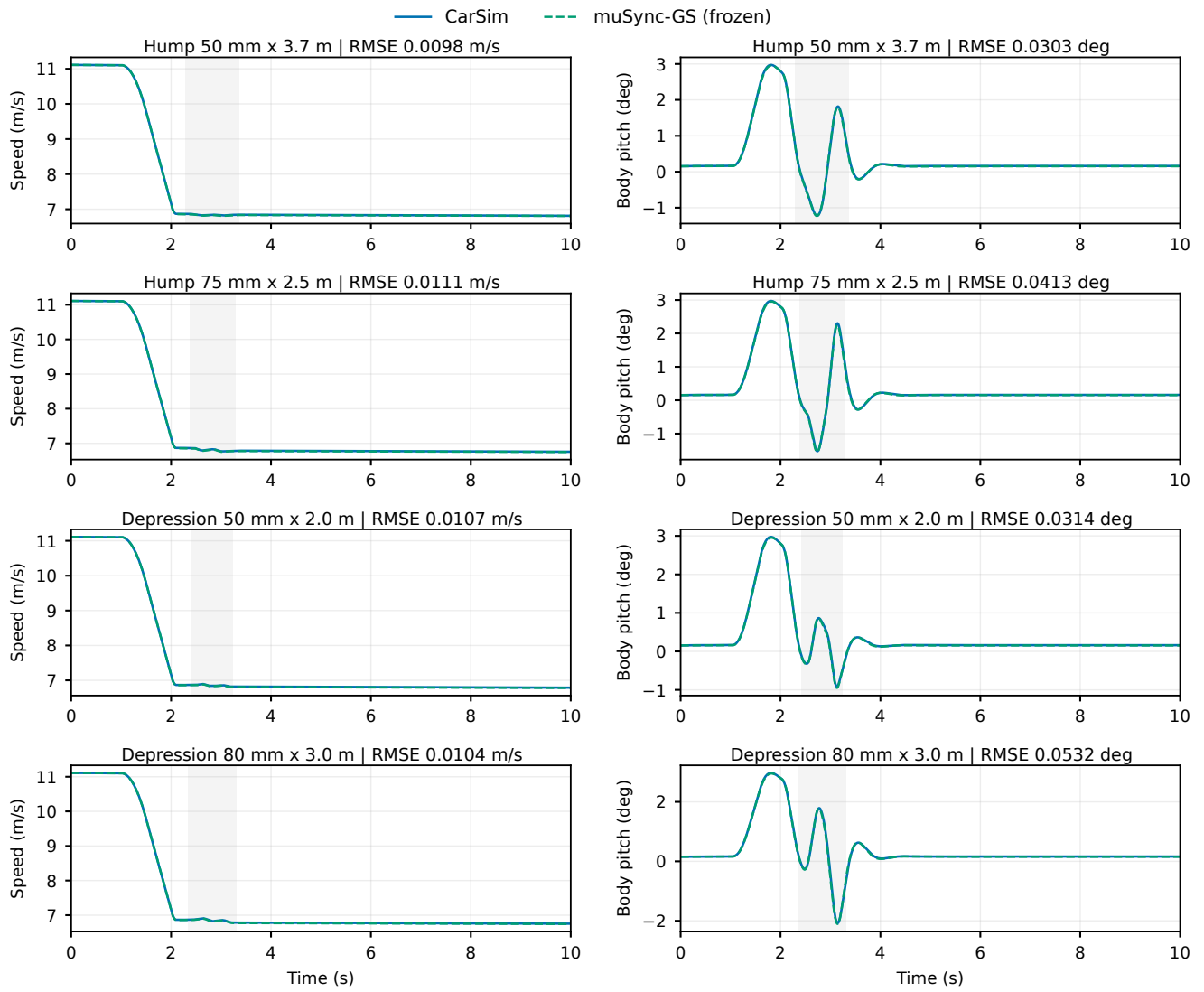


Figure 3: All four frozen road-elevation-profile holdouts. New feature heights, depths, lengths, and slopes are evaluated with the calibration frozen.

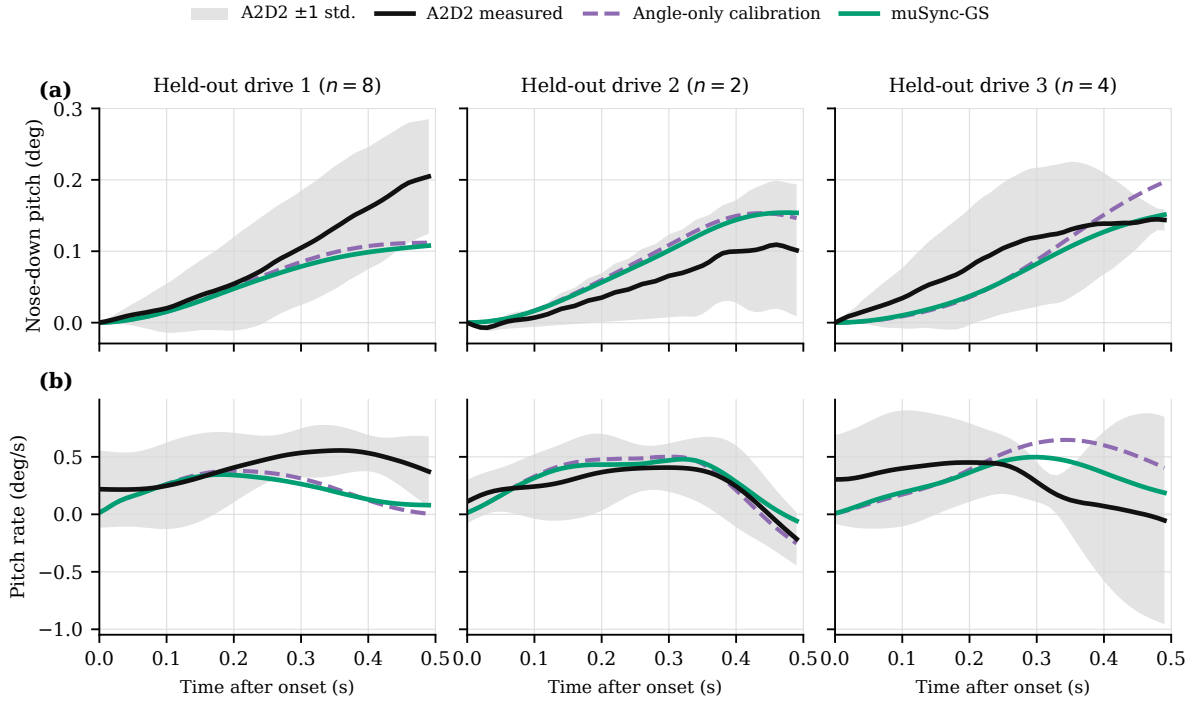


Figure 4: Complete three-fold leave-one-drive-out A2D2 brake-dive responses. Columns are held-out drives ( $n = 8, 2, 4$ ); rows show pitch angle and pitch rate. Shading is  $\pm 1$  event standard deviation, the dashed curve is the angle-only ablation, and green is joint pitch-rate vehicle adaptation. Calibration uses only the other two drives; held-out inference uses brake pressure and event-start speed.

## D Additional Video and Visual Results

### D.1 Video-observed motion curves

Figure 5 visualizes the CarSim-referenced weather- and geometry-response measurements summarized numerically in the main paper. The heavy-weather panels plot weather-minus-sunny normalized progress. The geometry panels plot event-local pitch after the fixed baseline and sign conver-

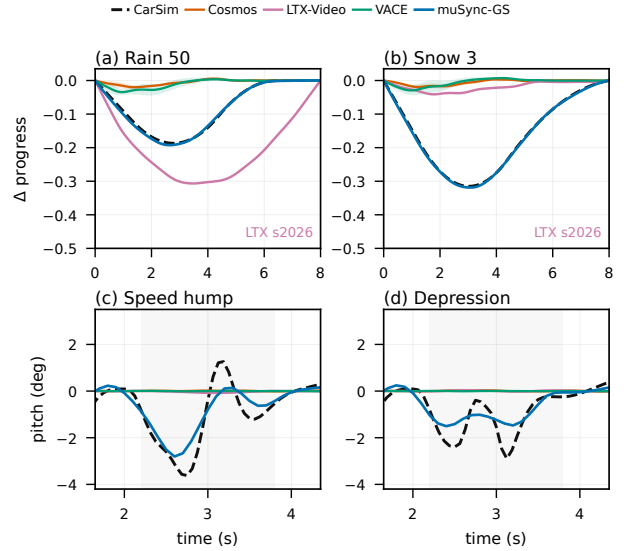


Figure 5: VGGT-recovered motion against CarSim. (a)–(b) Weather-minus-sunny normalized progress for Rain-50 and Snow-3. Baseline curves are three-seed means with bands; the labeled LTX seed shows its variation. (c)–(d) Event-local pitch for the speed hump and road depression. Gray shading marks the fixed 2.2–3.8 s event window.

sion defined in Appendix A.5. Cosmos and VACE remain near the sunny motion, as encouraged by the W preservation prompt, while the displayed LTX seed illustrates its larger seed-dependent deviation. These curves are not a physics-generation ranking. For geometry, near-zero baseline curves indicate little recovered pitch event, while muSync-GS follows the localized CarSim shape; profile differences still make this a semantic diagnostic. The rendered CarSim state reference is reported numerically in the main table and is omitted from this compact plot for legibility.

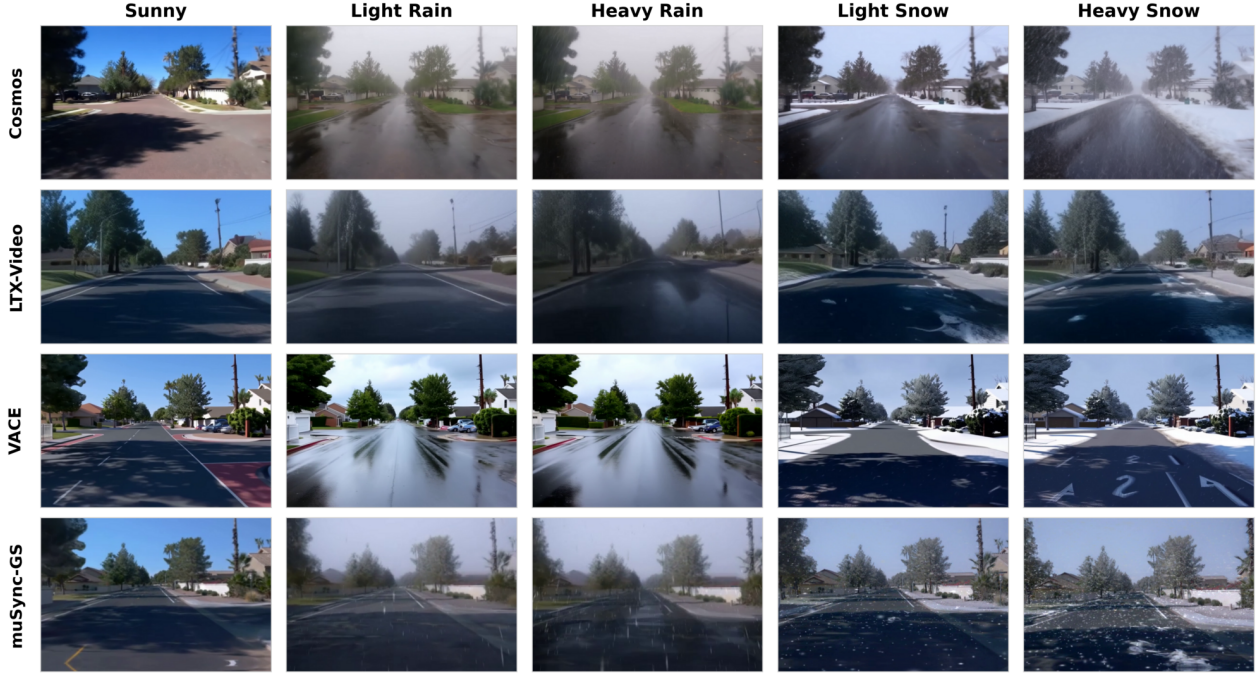
### D.2 Full weather-severity comparison

Figure 6 expands the three-condition main-paper figure to sunny, light rain, heavy rain, light snow, and heavy snow at the same source scene and frame. The light/heavy conditions are Rain-10/Rain-50 and Snow-1/Snow-3, respectively. This figure evaluates appearance control only; condition-dependent motion is reported separately as the prompt-constrained W diagnostic.

### D.3 Multi-scene qualitative consistency

The main quantitative video protocols use Waymo context 9385013624094020582 over 2547.650–2567.650 s, denoted scene 938501. The five segments below are additional qualitative scenes. Figure 7 extends the weather visualization to five distinct reconstructed Waymo segments (Sun et al. 2020). All selected clips are static or quasi-static and contain no nearby interaction-critical agent whose trajectory would require counterfactual resimulation. The columns correspond to scene IDs 117240, 148697, 144248, 150623, and 322492,

## Weather Controllability



Matched input scene and video time (frame 40, seed 2025). Quantitative scene and camera-path preservation are reported separately.

Figure 6: Five-level weather controllability at frame 40 and seed 2025. Every row uses a matched source scene and video time.

rather than multiple viewpoints from one route. Each column uses the same scene-specific rendering assets across the three weather conditions.

### D.4 Automatic quality and severity metrics

The primary automatic evaluation contains 128 videos. The weather subset has 108 videos: each of the four methods uses nine conditions and three visual seeds. The road subset has 20 videos: two tasks and three seeds per baseline (18), plus two deterministic muSync-GS outputs. For muSync-GS weather, only visual stochastic variables change across seeds; the physical state and camera trajectory are frozen. DOVER, FAST-VQA, CLIP severity, and temporal LPIPS use the 108-video weather subset; localization, anchor, and T-IoU use the 20-video road subset. Every video is normalized to H.264,  $480 \times 320$ , 10 fps, and 81 frames. The first 81 source frames are preserved and retimed, avoiding frame loss in the original 16-fps geometry outputs. Preserving all frames changes the playback duration of those baseline geometry clips; these no-reference and visual metrics are therefore clip-normalization diagnostics, not physical-time response measures.

DOVER (Wu et al. 2023) is the primary no-reference video-quality metric; FAST-VQA-B (Wu et al. 2022) is a robustness check with overlapping scope. Both use the authors’ official pretrained checkpoints and score-fusion code. For semantic weather severity, we uniformly sample 16 frames and

encode them with CLIP ViT-B/32 (Radford et al. 2021). Four-prompt ensembles describe dry, heavy-rain, and heavy-snow dashcam scenes. The exact dry prompts are “a clear dry road seen from a dashcam,” “a sunny driving scene with no precipitation,” “a dry suburban road viewed through a windshield,” and “clear weather in an on-board driving video.” The rain prompts are “a dashcam driving scene in heavy rain,” “a road seen through a windshield during heavy rainfall,” “an on-board driving video with intense rain,” and “a wet road in a severe rainstorm.” The snow prompts are “a dashcam driving scene in heavy snow,” “a road seen through a windshield during heavy snowfall,” “an on-board driving video with intense snow,” and “a snow-covered road in a severe snowstorm.” With normalized frame feature  $f_t$  and mean text feature  $g_w$ , the rain score is

$$s_{\text{rain}} = \frac{1}{16} \sum_{t=1}^{16} (f_t^\top g_{\text{rain}} - f_t^\top g_{\text{dry}}), \quad (21)$$

and snow is analogous. We report Spearman’s  $\rho$  between numeric input and severity score over sunny plus four positive levels. Adjacent accuracy is the fraction of consecutive pairs whose score increases. These are semantic control metrics, not photorealism measures.

Table 10 reports the individual muSync-GS seed results. The low but nonzero variation in VGGT and no-reference quality metrics is induced only by particles and spatial snow





Figure 7: Weather effects across five distinct reconstructed Waymo scenes. Rows show Sunny, Rain-50, and Snow-3; columns show the scene IDs. Representative frames are sampled at the same video index within each scene.

realization; CLIP severity ordering is unchanged. VACE obtains the highest DOVER and FAST-VQA values, while muSync-GS obtains perfect rain/snow severity ordering and the lowest road-anchor error. Together, these metrics distinguish perceptual quality from continuous severity control and localized road consistency.

### D.5 Flow-aligned temporal LPIPS diagnostic

We additionally sample 40 adjacent frame pairs per weather video. Farneback backward flow warps the preceding frame to the current frame, and LPIPS-Alex (Zhang et al. 2018) is evaluated in two fixed roadside regions, (0, 60, 150, 240) and (330, 60, 480, 240). The matching method/seed sunny score is subtracted to obtain excess temporal LPIPS.

This diagnostic is excluded from the main comparison because rain streaks and snow particles legitimately cross static-scene regions. It therefore combines intended moving precipitation with background flicker in a single score.

### D.6 Interpretation of the visual metrics

Together with the qualitative comparison, the road-geometry metrics characterize edit localization, temporal persistence, and the associated vehicle response. A high target-localized edit rate indicates that a change appears in the intended region. Anchor error and T-IoU add spatial and temporal evidence, while the geometry-response evaluation tests whether the visible feature is accompanied by the expected vehicle response. The scene-preservation metric is progress aligned so that differences in braking duration are not counted as lateral camera drift.



| (a) Aggregate no-reference weather-video quality |                |               |               |               |  |
|--------------------------------------------------|----------------|---------------|---------------|---------------|--|
| Method                                           | Tech. raw      | Aesth. raw    | DOVER Overall | FAST          |  |
| Cosmos                                           | -0.1361        | -0.0376       | 0.1760        | 0.4134        |  |
| LTX-Video                                        | -0.1335        | -0.0233       | 0.2033        | 0.4014        |  |
| VACE                                             | <b>-0.1009</b> | <b>0.0345</b> | <b>0.3679</b> | <b>0.6514</b> |  |
| muSync-GS                                        | -0.1274        | -0.0476       | 0.1800        | 0.4545        |  |

| (b) Severity ordering and road-anchor consistency |               |               |               |               |                     |
|---------------------------------------------------|---------------|---------------|---------------|---------------|---------------------|
| Method                                            | Rain $\rho$   | Snow $\rho$   | Rain Adj.     | Snow Adj.     | Anchor $\downarrow$ |
| Cosmos                                            | 0.9667        | 0.8667        | 0.9167        | 0.6667        | 0.678               |
| LTX-Video                                         | 0.9333        | 0.8667        | 0.8333        | 0.8333        | 0.708               |
| VACE                                              | 0.9333        | 0.6667        | 0.8333        | 0.7500        | 0.440               |
| muSync-GS                                         | <b>1.0000</b> | <b>1.0000</b> | <b>1.0000</b> | <b>1.0000</b> | <b>0.296</b>        |

| (c) Flow-aligned temporal LPIPS over positive weather levels |          |             |          |             |  |
|--------------------------------------------------------------|----------|-------------|----------|-------------|--|
| Method                                                       | Rain raw | Rain excess | Snow raw | Snow excess |  |
| Cosmos                                                       | 0.0438   | 0.0002      | 0.0447   | 0.0012      |  |
| LTX-Video                                                    | 0.0385   | -0.0001     | 0.0522   | 0.0136      |  |
| VACE                                                         | 0.0451   | 0.0004      | 0.0538   | 0.0092      |  |
| muSync-GS                                                    | 0.0734   | 0.0510      | 0.1482   | 0.1258      |  |

Table 9: Automatic visual evaluation. Technical and Aesthetic are raw DOVER fields; Overall is the fused DOVER score; FAST is FAST-VQA-B; Adj. is adjacent-level ordering accuracy; Anchor is normalized road-anchor error. Panels (a), the weather columns of (b), and (c) use the 108-video weather subset; Anchor uses the 20-video road subset. The main paper reports three-seed mean  $\pm$  sample SD for its selected columns, and Table 10 gives the individual muSync-GS seeds. The muSync-GS road output is deterministic. Flow-aligned roadside temporal LPIPS is diagnostic because visible precipitation also increases the score.

| (a) Recovered-motion errors |                         |                           |
|-----------------------------|-------------------------|---------------------------|
| Seed                        | Weather diagnostic RMSE | Lateral RMSE $\downarrow$ |
| 2025                        | 0.004241                | 0.002600                  |
| 2026                        | 0.004345                | 0.002732                  |
| 2027                        | 0.004146                | 0.002653                  |
| Mean $\pm$ SD               | 0.004244 $\pm$ 0.000099 | 0.002662 $\pm$ 0.000067   |

| (b) No-reference quality |                          |                     |
|--------------------------|--------------------------|---------------------|
| Seed                     | DOVER Overall $\uparrow$ | FAST $\uparrow$     |
| 2025                     | 0.1787                   | 0.4542              |
| 2026                     | 0.1825                   | 0.4547              |
| 2027                     | 0.1788                   | 0.4546              |
| Mean $\pm$ SD            | 0.1800 $\pm$ 0.0021      | 0.4545 $\pm$ 0.0003 |

| (c) Severity rank correlation |                   |                   |
|-------------------------------|-------------------|-------------------|
| Seed                          | Rain $\rho$       | Snow $\rho$       |
| 2025                          | 1.000             | 1.000             |
| 2026                          | 1.000             | 1.000             |
| 2027                          | 1.000             | 1.000             |
| Mean $\pm$ SD                 | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 |

Table 10: Per-seed muSync-GS weather robustness. Weather RMSE is the VGGT progress-response error; lateral RMSE is progress-aligned lateral RMS; DOVER Overall and FAST are macro-averages over nine weather conditions. Only visual stochastic variables change. Lateral RMSE first macro-averages the eight non-sunny conditions within each seed, then reports the mean and sample standard deviation across the three seeds.

## E Bounded Surface-Parameter Sensitivity

### E.1 Scope of the analysis

This analysis evaluates the reported weather–friction ordering across bounded perturbations of the dense-asphalt parameter vector. Precipitation remains a deterministic input, while the listed surface parameters are sampled over the declared engineering ranges. Literature and standards provide the physical scale or measurement basis for the selected nominal values.

### E.2 Surface equations and parameter basis

For rain, let  $I_r$  be rainfall intensity in mm/h, MTD be mean texture depth in mm,  $L_d$  drainage-path length in m, and  $S$  crossfall in m/m. The metric Gallaway relation gives

$$\begin{aligned} h_t &= 0.01485 \text{MTD}^{0.11} L_d^{0.43} I_r^{0.59} S^{-0.42}, \\ h_f &= \max(0, h_t - \text{MTD}). \end{aligned} \quad (22)$$

Here  $h_t$  is the total water depth, in millimeters, measured from the bottom of the pavement macrotexture and used in the main-paper surface state. The derived  $h_f$ , also in millimeters, is the nonnegative free-water depth above the texture peaks and is retained only as a diagnostic. The reported rain-friction mapping uses  $h_t$  rather than treating  $h_f$  as an additional fitted friction input. The variables and exponents originate from controlled pavement-drainage experiments (Gallaway, Schiller, and Rose 1971; Gallaway et al. 1979). The raw rain-condition coefficient is

$$\begin{aligned} \mu_c^{\text{rain}} &= \mu_{\text{dry}} \left[ r_f + (1 - r_f) e^{-h_t/h_c} \right], \\ r_f &= \mu_{\text{flooded}} / \mu_{\text{dry}}, \end{aligned} \quad (23)$$

where  $h_c$  is a film-decay scale in millimeters, applied to the same bottom-referenced water-depth convention. For water-equivalent snowfall  $I_s$  accumulated for duration  $T$ ,

$$\begin{aligned} h_s &= I_s T \rho_w / \rho_s, \\ c_s &= \min(1, h_s / h_{\text{cover}}), \\ \mu_c^{\text{snow}} &= \mu_{\text{bw}} - (\mu_{\text{bw}} - \mu_{\text{snow}}) c_s. \end{aligned} \quad (24)$$

Here  $\rho_w = 1000 \text{ kg/m}^3$  is the water density and  $\rho_s$  is the fresh-snow density;  $c_s$  is a contact-coverage proxy, and  $h_{\text{cover}}$  is a model threshold rather than a road-design standard.

Texture depth and crossfall support the dense-asphalt nominal, while drainage length comes from the selected route. Film decay and full-cover depth are explicit model transition scales.

### E.3 Sampling and frozen simulation protocol

We generate 128 independent Latin-hypercube samples for each weather family with seed 20260725. Rain samples  $\{\mu_{\text{dry}}, \mu_{\text{flooded}}, \text{MTD}, L_d, S, h_c\}$ , and snow samples  $\{\mu_{\text{dry}}, \mu_{\text{bw}}, \mu_{\text{snow}}, \rho_s, h_{\text{cover}}\}$ , uniformly within the corresponding bounds in Table 11. Rain uses 21 intensities from 0 to 50 mm/h; snow uses 13 water-equivalent rates from 0 to 3 mm/h with  $T = 1 \text{ h}$ . Dense nominal curves use 51 and 31 rates, respectively. The experiment contains 4,434 solver cases.

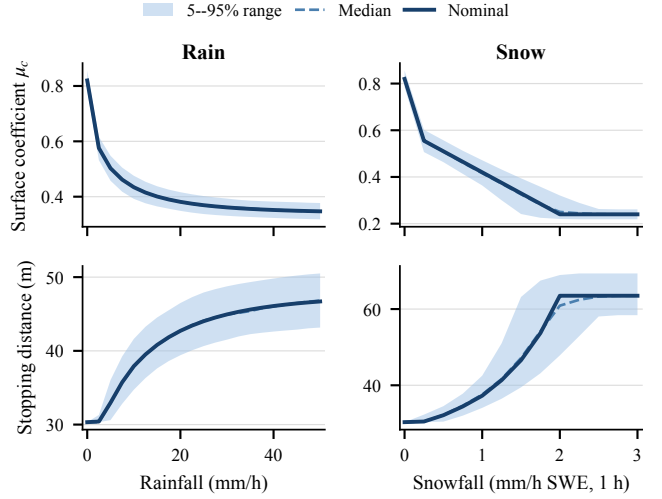


Figure 8: Bounded surface-parameter sensitivity under the frozen brake/ABS protocol. Top: raw condition coefficient  $\mu_c$ . Bottom: stopping distance. Solid lines use nominal parameters; bands are the 5th–95th percentiles over the 128 sampled parameter sets.

All non-surface quantities are frozen to the common braking protocol: initial speed 60 km/h, total brake torque 2300 N m, front bias 0.65, torque ramp 0.5 s, ABS release width 0.2, and internal step  $5 \times 10^{-4} \text{ s}$ . Each raw coefficient passes through the frozen family-specific tire efficiency and slip-stiffness response terms before the same brake solver is run. Against archived nominal outputs, the maximum raw- $\mu$  discrepancy is  $4.3 \times 10^{-11}$  and the maximum stopping-distance discrepancy is 1.37 mm.

### E.4 Envelope and rank-sensitivity results

All sampled curves are monotonic over strictly positive precipitation rates; the exact zero-rate point invokes the dry tire-family branch and is excluded from that statement. At Rain-50, nominal raw  $\mu_c$  is 0.347 with a 0.318–0.377 envelope, and stopping distance is 46.70 m with a 43.16–50.50 m envelope. Stopping-distance envelopes are 34.07–42.53 m at Snow-1 and 58.39–69.35 m at saturated Snow-3.

Spearman correlations with stopping distance at Rain-50 are dominated by  $\mu_{\text{flooded}}$  ( $\rho = -0.973$ ). At Snow-1, the largest absolute correlations are full-cover depth ( $-0.601$ ), snow density ( $-0.535$ ), bare-wet friction ( $-0.489$ ), and snow friction ( $-0.354$ ). At Snow-3, full coverage is saturated and the snow coefficient dominates ( $\rho = -1.000$ ). These rankings match the model equations; partial coverage depends on accumulation and transition depth, whereas saturated coverage depends mainly on the snow-contact friction anchor.

| Parameter                                  | Nominal               | Sampled bound               | Source basis and interpretation                                                                                                                                                                                                            |
|--------------------------------------------|-----------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dry coefficient $\mu_{\text{dry}}$         | 0.82                  | [0.78, 0.86]                | Frozen anchor with an approximately $\pm 5\%$ perturbation. The nominal is inside the 0.8–1.0 dry-bare friction-number range summarized by VTI (Wallman and Åström 2001, p. 36).                                                           |
| Flooded coefficient $\mu_{\text{flooded}}$ | 0.34                  | [0.306, 0.374]              | Frozen anchor with $\pm 10\%$ perturbation. Wet-road measurements show a broad surface-dependent range (Ivan et al. 2010).                                                                                                                 |
| Texture depth MTD                          | 0.50 mm               | [0.40, 0.60] mm             | Dense-graded-asphalt nominal with $\pm 20\%$ perturbation. Field MPD statistics have a comparable scale (Boz, Flintsch, and de León Izeppi 2023). Volumetric MTD measurement is standardized by ASTM E965/E965M (ASTM International 2019). |
| Drainage length $L_d$                      | 3.75 m                | [3.0, 4.5] m                | Scenario flow-path nominal with $\pm 20\%$ perturbation; $L_d$ is the route-dependent input to the Gallaway model (Gallaway, Schiller, and Rose 1971).                                                                                     |
| Crossfall $S$                              | 2.0%                  | [1.6, 2.4]%                 | $\pm 20\%$ around the nominal. FHWA HEC-22 lists 1.5–2.0% for high-type two-lane surfaces and notes that 2.5% may aid drainage in intense-rainfall regions (Kilgore, Atayee, and Herrmann 2024).                                           |
| Film-decay scale $h_c$                     | 0.30 mm               | [0.24, 0.36] mm             | Model-specific transition scale with $\pm 20\%$ perturbation.                                                                                                                                                                              |
| Fresh-snow density $\rho_s$                | 100 kg/m <sup>3</sup> | [80, 120] kg/m <sup>3</sup> | $\pm 20\%$ around the conventional 10:1 water-to-snow nominal. Four-year observations report site means of 72–103 kg/m <sup>3</sup> and wider event variability (Judson and Doesken 2000).                                                 |
| Full-cover threshold $h_{\text{cover}}$    | 20 mm                 | [16, 24] mm                 | Model-specific tire-contact interpolation threshold with $\pm 20\%$ perturbation.                                                                                                                                                          |
| Bare-wet transition $\mu_{\text{bw}}$      | 0.60                  | [0.54, 0.66]                | Frozen transition anchor with $\pm 10\%$ perturbation. It lies within broad field wet-pavement measurements (Ivan et al. 2010).                                                                                                            |
| Snow coefficient $\mu_{\text{snow}}$       | 0.24                  | [0.216, 0.264]              | $\pm 10\%$ around the fresh-snow anchor, consistent with the 0.20–0.25 new-snow skid-resistance range in controlled tests (Ichihara and Mizoguchi 1970).                                                                                   |

Table 11: Nominal surface parameters and declared sensitivity bounds. Intervals are controlled perturbations unless explicitly stated otherwise. MPD and volumetric MTD are related macrotexture measures but are not identical.

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